

CODE SPORT



In this month's column, we discuss a few coding questions.

In last month's column, we had discussed some coding interview questions on arrays, heaps and sequences. Some of our readers have asked me to continue this discussion and hence, in this month's column, we consider a few more coding interview questions.

Instead of providing solutions to the questions, I will just present the questions for our readers to practice and we will discuss the solutions in next month's column.

When you are solving these problems, the one thing to keep in mind is to identify whether there is a common underlying pattern. Knowing how to solve one specific pattern will often help you to apply it to multiple specific problems.

Recall this question that was featured in last month's column: We were given a set of buildings on a street and we were asked to find the buildings from which we can see the sunset. Essentially, for each building we had to find whether there were any taller buildings blocking its view towards the west.

A related problem is the stock span problem. You are given the daily price of a stock for N days. You are asked to find the stock span for each day.

Stock span is defined as the number of consecutive days from the given day, for which the price of the stock is less than or equal to the price on the given day. Given an array of stock prices {98, 82, 59, 72, 60, 75, 86}, the stock span for each of the seven days would be {1, 1, 1, 2, 1, 4, 6}.

On a simple first glance of the description, the stock span problem and the building height problem look totally unrelated. But the key is to discard the description of the situation/context and just look at the mathematical model of the problem.

In both these problems, for each array element at index i , we need to find the previous greater element at an index j such that $j < i$. In the first case, if there is a previous greater element, then the current building cannot view the sunset. In the stock span problem, the span ends just before the index of the previous greater element. Even though the textual description of both these problems looks completely different, the underlying pattern for both of them is to find the previous greater element for each array element. Once you understand how to solve the problem of determining the previous greater element, we can solve both the above problems.

So now our problem is this: Given an array of N integers, for each array element at index i , find the index of its previous greater element. We can consider index $i=0$ (arrays are indexed starting at 0) as a special case and consider that it has no Previous Greater Element (PGE), and we will initialise its PGE to be -1 (since there are no elements before it). How do we now compute PGE for the remaining array indices?

The simplest approach is to scan backwards from each index i , terminating either at the first element we see which is greater than $A[i]$, or when we hit the beginning of the array.

